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§23.18 Modular Transformations

Elliptic Modular Function

$\lambda(\mathscr{A}\tau)$ equals

23.18.1
$$\begin{gathered}\lambda(\tau),\\ 1-\lambda(\tau),\\ \frac{1}{\lambda(\tau)},\\ \frac{1}{1-\lambda(\tau)},\\ \frac{\lambda(\tau)}{\lambda(\tau)-1},\\ 1-\frac{1}{\lambda(\tau)},\end{gathered}$$

according as the elements $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ of $\mathscr{A}$ in (23.15.3) have the respective forms

23.18.2
$$\begin{gathered}\begin{bmatrix} \mathrm{o} & \mathrm{e} \\ \mathrm{e} & \mathrm{o} \end{bmatrix},\\ \begin{bmatrix} \mathrm{e} & \mathrm{o} \\ \mathrm{o} & \mathrm{e} \end{bmatrix},\\ \begin{bmatrix} \mathrm{o} & \mathrm{e} \\ \mathrm{o} & \mathrm{o} \end{bmatrix},\\ \begin{bmatrix} \mathrm{e} & \mathrm{o} \\ \mathrm{o} & \mathrm{o} \end{bmatrix},\\ \begin{bmatrix} \mathrm{o} & \mathrm{o} \\ \mathrm{e} & \mathrm{o} \end{bmatrix},\\ \begin{bmatrix} \mathrm{o} & \mathrm{o} \\ \mathrm{o} & \mathrm{e} \end{bmatrix}.\end{gathered}$$

Here e and o are generic symbols for even and odd integers, respectively. In particular, if $a-1, b, c$, and $d-1$ are all even, then

23.18.3
$$\lambda(\mathscr{A}\tau) = \lambda(\tau),$$

and $\lambda(\tau)$ is a cusp form of level zero for the corresponding subgroup of $\mathrm{SL}(2,\mathbb{Z})$.

Klein's Complete Invariant

23.18.4
$$J(\mathscr{A}\tau) = J(\tau).$$

$J(\tau)$ is a modular form of level zero for $\mathrm{SL}(2,\mathbb{Z})$.

Dedekind's Eta Function

23.18.5
$$\eta(\mathscr{A}\tau) = \varepsilon(\mathscr{A})\big(-\mathrm{i}(c\tau+d)\big)^{1/2}\eta(\tau),$$

where the square root has its principal value and

23.18.6 $$\varepsilon(\mathcal{A}) = \exp\left(\pi\mathrm{i}\left(\frac{a+d}{12c} + s(-d,c)\right)\right),$$

23.18.7 $$s(d,c) = \sum_{r=1}^{c-1} \frac{r}{c}\left(\frac{dr}{c} - \left\lfloor\frac{dr}{c}\right\rfloor - \frac{1}{2}\right),$$

$c > 0$.

Here $s(d,c)$ is a Dedekind sum. See (27.14.11), §27.14(iii), §27.14(iv) and Apostol (1990, pp. 48 and 51–53). Note that $\eta(\tau)$ is of level $\frac{1}{2}$.

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